

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION
CLASS IX • SCIENCE (SUBJECT CODE: 086) • ACADEMIC SESSION 2026-27

SET E — CASE STUDY & PRACTICAL EXPERIMENTS

CHAPTER 05: EXPLORING MIXTURES: SOLUTION, COLLOID & SUSPENSION
TIME ALLOWED: 45 MINUTES MAXIMUM MARKS: 25 MARKS SERIES: CBS-IX-SCI-SET-E-05

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

GENERAL INSTRUCTIONS (SET E - CASE STUDY & PRACTICAL EXPERIMENTS):

1. The question paper comprises **5 In-Depth Practical Case Studies** with **5 MCQs each (Total: 25 Questions, 1 Mark each)**.
2. Read each laboratory experiment, data table, and passage carefully before answering the corresponding questions.
3. Each question has 4 options with a single correct answer.
4. All questions are compulsory. There is no negative marking.
5. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

CASE STUDY 1: COMPARATIVE OPTICAL & FILTRATION ANALYSIS: SOLUTION, COLLOID & SUSPENSION

Students were provided with three unlabeled beakers containing mixtures in water:

- **Beaker A:** 5 g of Copper Sulfate (CuSO_4) dissolved in 100 mL of water.
- **Beaker B:** 2 g of fine Starch powder boiled and cooled in 100 mL of water.
- **Beaker C:** 5 g of finely crushed Chalk powder stirred in 100 mL of water.

The students conducted three diagnostic practical tests:

1. **Tyndall Effect:** Directing a red laser beam through each beaker in a dark room.
2. **Filtration:** Passing each mixture through Whatman No. 1 filter paper.
3. **Stability:** Leaving the beakers undisturbed for 30 minutes.

Q.1 In which beaker(s) was the luminous path of the laser beam distinctly visible from the side (Tyndall effect)?

- (A) Beaker A only (B) Beaker B and Beaker C (while suspended)
(C) Beaker A and Beaker B only (D) None of the beakers

Q.2 What was retained on the filter paper after filtering the contents of Beaker C (Chalk and water)?

- (A) Zero residue; the mixture passed through completely (B) White insoluble chalk powder residue, while clear water passed as filtrate
(C) Blue copper sulfate crystals (D) A jelly-like starch sol

Q.3 What happened to the three mixtures after standing undisturbed for 30 minutes?

- (A) Beakers A and B remained completely homogeneous; Beaker C showed chalk settling to the bottom (B) All three mixtures settled completely to the bottom
(C) Beaker B separated into two distinct liquid layers (D) Beaker A evaporated completely

Q.4 How is Beaker B (Starch in water) accurately classified in colloid chemistry?

- (A) Gel (liquid in solid) (B) Sol (solid dispersed in liquid)
(C) Emulsion (liquid in liquid) (D) Aerosol (solid in gas)

Q.5 What microscopic kinetic phenomenon keeps colloidal particles in Beaker B from settling over time?

- (A) Continuous random zigzag Brownian motion caused by collisions with solvent molecules
(B) Magnetic attraction to the glass
(C) Gravitational repulsion
(D) Capillary ascent

CASE STUDY 2: SOLUBILITY CURVE & CRYSTALLIZATION OF COPPER SULFATE

A chemistry student investigated the temperature dependence of solubility of Copper Sulfate (CuSO_4) in water.

• **Solubility Data (g of CuSO_4 per 100 g of water):**

- At 20°C : 21.0 g
- At 40°C : 29.0 g
- At 60°C : 40.0 g
- At 80°C : 55.0 g

The student prepared a saturated solution by dissolving 55.0 g of CuSO_4 in 100.0 g of water at 80°C . She filtered the hot solution to remove insoluble impurities and allowed the clear filtrate to cool slowly undisturbed to 20°C .

Q.6 What mass of solid Copper Sulfate crystals precipitates out of solution as it cools from 80°C to 20°C ?

- (A) 21.0 g
(B) 34.0 g
(C) 55.0 g
(D) 14.0 g

Q.7 Why is slow cooling preferred over rapid chilling with ice during crystallization?

- (A) To prevent the beaker from cracking
(B) Slow cooling allows well-formed, large, highly pure geometric crystals to grow without trapping solvent impurities
(C) To evaporate all the water
(D) To prevent chemical decomposition of copper

Q.8 Why is crystallization superior to simple evaporation to dryness for purifying salts?

- (A) Evaporation requires zero heat
(B) During evaporation to dryness, heat-sensitive compounds can decompose or char, and soluble impurities remain mixed with the dried solid
(C) Crystallization takes less time than evaporation
(D) Crystallization converts the salt into an element

Q.9 What is the mass percentage of CuSO_4 in the saturated solution at 20°C (21.0 g solute in 100.0 g water)?

- (A) 21.0%
(B) 17.36%
(C) 26.58%
(D) 19.0%

Q.10 What type of solution is formed immediately if a hot saturated solution at 80°C is cooled to 40°C without any crystal nucleation occurring?

- (A) Unsaturated solution
(B) Supersaturated solution
(C) Dilute solution
(D) Colloidal suspension

CASE STUDY 3: PAPER CHROMATOGRAPHY OF SYNTHETIC BLACK INK

In a forensic chemistry laboratory practical, students analyzed water-soluble black marker ink using ascending paper chromatography.

1. A pencil baseline was drawn 2 cm from the bottom edge of a Whatman chromatography paper strip.

2. A small concentrated spot of black ink was applied on the baseline using a capillary tube and allowed to dry.
3. The strip was suspended vertically in a chromatography jar containing water as the mobile solvent, with the solvent level 1 cm below the pencil line.
4. As the solvent ascended by capillary action, the single black spot separated into three distinct color bands: Yellow, Red, and Blue.
5. When the solvent front reached 10.0 cm from the baseline, the strip was removed and dried:
 - Yellow dye travelled: 3.0 cm
 - Red dye travelled: 6.0 cm
 - Blue dye travelled: 8.5 cm

Q.11 Why was the reference baseline drawn with a graphite pencil rather than an ordinary ballpoint ink pen?

- | | |
|---|--|
| (A) Pencil graphite is insoluble in the running mobile solvent (water) and will not dissolve or interfere with the chromatogram | (B) Pencil graphite conducts electricity |
| (C) Pencil lines are darker than ink | (D) Pencil reacts chemically with paper |

Q.12 Calculate the Retention Factor (R_f) value for the Red dye:

- | | |
|----------|----------|
| (A) 0.30 | (B) 0.60 |
| (C) 0.85 | (D) 1.67 |

Q.13 Which of the three dye components had the highest solubility in the mobile solvent (water) and lowest adsorption to the cellulose paper?

- | | |
|----------------|--|
| (A) Yellow dye | (B) Red dye |
| (C) Blue dye | (D) All three had identical solubility |

Q.14 Why must the solvent level in the developing chamber be kept below the pencil baseline spot?

- | | |
|---|---|
| (A) To prevent the paper from getting wet | (B) To prevent the ink spot from directly dissolving into the reservoir solvent before capillary migration begins |
| (C) To allow air bubbles to escape | (D) To keep the chamber at low temperature |

Q.15 Which of the following mixtures cannot be separated using paper chromatography?

- | | |
|--|---|
| (A) Amino acids in a protein hydrolysate | (B) Plant leaf pigments (chlorophyll a, chlorophyll b, xanthophyll, carotene) |
| (C) Dyes in confectionery food coloring | (D) A mixture of sand and iron filings |

CASE STUDY 4: EXTRACTION OF IMMISCIBLE LIQUIDS USING A SEPARATING FUNNEL

In a separation science experiment, a student was tasked with separating a 150 mL mixture of kerosene oil and water.

- Density of kerosene oil = 0.80 g/cm^3
- Density of pure water = 1.00 g/cm^3

The student poured the heterogeneous mixture into a glass separating funnel supported on a ring stand, stoppered it, inverted and vented it, and then left it undisturbed for 15 minutes.

Two distinct transparent layers formed with a sharp horizontal interface.

She carefully opened the stopcock to collect the lower liquid layer into Beaker 1 and closed it just as the interface touched the stopcock, then poured the upper liquid layer out through the top neck into Beaker 2.

Q.16 Which liquid formed the lower layer in the separating funnel and was collected first in Beaker 1?

- (A) Kerosene oil, because it is lighter (B) Water, because it has higher density ($1.00 > 0.80 \text{ g/cm}^3$)
(C) An emulsion of both liquids (D) Air bubbles

Q.17 Why is the upper liquid layer (kerosene) poured out through the top opening rather than drained through the bottom stopcock?

- (A) To prevent the stopcock from rusting (B) To avoid contaminating the purified upper liquid with droplets of the denser lower liquid clinging to the stopcock bore
(C) Because kerosene cannot flow through a narrow nozzle (D) To allow kerosene to evaporate faster

Q.18 Why was the student instructed to invert the stoppered separating funnel and open the stopcock briefly (venting)?

- (A) To release internal vapor pressure built up during shaking (B) To allow external air to react chemically with kerosene
(C) To stir the mixture thoroughly (D) To clean the nozzle

Q.19 Which physical property is the fundamental basis for separation using a separating funnel?

- (A) Difference in boiling points (B) Difference in solubilities in alcohol
(C) Immiscibility and difference in liquid densities (D) Difference in magnetic properties

Q.20 Which of the following liquid pairs can be separated using a separating funnel?

- (A) Ethanol and water (B) Mustard oil and water
(C) Acetone and water (D) Acetic acid and water

CASE STUDY 5: MIXTURE VS COMPOUND: HEATING IRON FILINGS AND SULFUR POWDER

Students investigated the difference between a physical mixture and a chemical compound using iron filings and sulfur powder in a mass ratio of 7:4.

- **Sample 1 (Mixture):** Iron filings and sulfur powder were thoroughly ground together in a mortar at room temperature.
- **Sample 2 (Compound):** An identical mixture was placed in a hard glass test tube and heated strongly over a Bunsen flame. A reddish glow spread through the mass, forming a hard black crystalline lump.

The students tested both samples with a bar magnet, Carbon Disulfide (CS_2) solvent, and dilute Hydrochloric Acid (HCl).

Q.21 What happened when a bar magnet was brought near Sample 1 (Mixture) and Sample 2 (Compound)?

- (A) Both samples were attracted completely to the magnet (B) In Sample 1, iron filings were attracted to the magnet, leaving yellow sulfur behind; Sample 2 was not attracted to the magnet
(C) Neither sample was attracted to the magnet (D) Sample 2 was repelled by the magnet with violent sparks

Q.22 What was observed when Carbon Disulfide (CS_2) solvent was added to Sample 1?

- (A) Iron dissolved, leaving sulfur (B) Yellow sulfur dissolved completely, leaving grey iron filings behind; upon evaporation of CS_2 , yellow sulfur crystals reformed
(C) The mixture exploded (D) Zero dissolution occurred

Q.23 What gas was evolved when dilute HCl was added to Sample 1 (Mixture of Fe and S)?

- (A) Sulfur dioxide gas (SO_2) (B) Colorless, odorless Hydrogen gas (H_2) that burns with a 'pop' sound
- (C) Hydrogen sulfide gas (H_2S) smelling like rotten eggs (D) Chlorine gas (Cl_2)

Q.24 What gas was evolved when dilute HCl was added to Sample 2 (Compound FeS)?

- (A) Hydrogen gas (H_2) (B) Colorless Hydrogen Sulfide gas (H_2S) having a characteristic foul smell of rotten eggs
- (C) Oxygen gas (O_2) (D) Carbon dioxide gas (CO_2)

Q.25 Why is the formation of Iron Sulfide from iron and sulfur classified as an exothermic chemical change?

- (A) Because it can be easily reversed using a magnet (B) Because new chemical bonds form with release of energy (red-hot glow) and the product FeS has properties entirely different from iron and sulfur
- (C) Because no heat was involved (D) Because the mass decreased by 50%

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET E)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & PRACTICAL INVESTIGATION EXPLANATIONS (SET E)

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.1	(B)	Colloidal particles in Beaker B (starch sol, 1–100 nm) and suspended particles in Beaker C scatter visible light. True solutions in Beaker A (< 1 nm) do not show Tyndall effect.
Q.2	(B)	Chalk particles in a suspension are larger than 100 nm and cannot pass through the microscopic pores of filter paper, leaving solid residue.
Q.3	(A)	True solutions (Beaker A) and colloids (Beaker B) are stable and do not settle; suspensions (Beaker C) are unstable and settle under gravity.
Q.4	(B)	Starch powder consists of solid macromolecules dispersed in a liquid dispersion medium (water), which defines a sol.
Q.5	(A)	Erratic, continuous collisions between colloidal starch particles and rapidly moving water molecules (Brownian motion) counteract gravitational sedimentation.
Q.6	(B)	Mass precipitated $= \text{Solubility at } 80^{\circ}\text{C} - \text{Solubility at } 20^{\circ}\text{C} = 55.0\text{ g} - 21.0\text{ g} = 34.0\text{ g}$.
Q.7	(B)	Slow, undisturbed cooling allows solute molecules to arrange methodically into a pure crystal lattice, excluding foreign impurities.
Q.8	(B)	Evaporation leaves all soluble contaminants behind with the dried product; crystallization purifies the compound by selectively forming pure crystal lattices.
Q.9	(B)	$\text{Mass percentage} = \frac{\text{Mass of Solute}}{\text{Mass of Solution}} \times 100 = \frac{21.0}{21.0 + 100.0} \times 100 = \frac{21.0}{121.0} \times 100 \approx 17.36\%$.
Q.10	(B)	A solution that temporarily holds more dissolved solute than its thermodynamic saturation equilibrium at that temperature is supersaturated.
Q.11	(A)	Graphite is insoluble in water and organic solvents; ink from a pen would dissolve and migrate, ruining the chromatographic baseline.
Q.12	(B)	$R_f = \frac{\text{Distance travelled by solute component}}{\text{Distance travelled by solvent front}} = \frac{6.0\text{ cm}}{10.0\text{ cm}} = 0.60$.
Q.13	(C)	The Blue dye migrated the farthest (\$8.5 cm, $R_f = 0.85$), indicating greatest solubility in mobile water and weakest affinity for stationary paper fibers.
Q.14	(B)	If submerged below the solvent surface, the ink spot simply washes out into the solvent pool instead of migrating up the paper strip.
Q.15	(D)	Sand and iron filings are macroscopic insoluble solids separated mechanically using a magnet; chromatography separates molecular solutes.
Q.16	(B)	The denser liquid (water, 1.00 g/cm^3) sinks to the bottom, while the less dense liquid (kerosene, 0.80 g/cm^3) floats on top.
Q.17	(B)	Draining the upper liquid through the bottom stopcock would wash residual droplets of the lower liquid into the clean beaker, causing cross-contamination.
Q.18	(A)	Shaking volatile liquids builds up internal vapor pressure that can pop the glass stopper; venting releases the pressure safely.
Q.19	(C)	Separating funnels are used exclusively for immiscible liquids that do not dissolve in each other and form distinct layers due to density differences.
Q.20	(B)	Mustard oil and water are completely immiscible and separate into two distinct layers; ethanol, acetone, and acetic acid are completely miscible with water.
Q.21	(B)	In a mixture, iron retains its magnetic properties. In the compound Iron Sulfide (FeS), iron is chemically bonded in a crystal lattice and loses its ferromagnetism.

Q. No.	Correct	Scientific Rationale & Experimental Data Analysis
Q.22	(B)	Sulfur is readily soluble in non-polar carbon disulfide (CS_2), while iron is insoluble. This is a physical separation method for the mixture.
Q.23	(B)	In the mixture, iron reacts independently with dilute acid: $Fe + 2HCl \rightarrow FeCl_2 + H_2 \uparrow$. Hydrogen is odorless and burns with a pop.
Q.24	(B)	Iron sulfide reacts chemically: $FeS + 2HCl \rightarrow FeCl_2 + H_2S \uparrow$. Hydrogen sulfide is toxic and smells like rotten eggs.
Q.25	(B)	Chemical reaction: $Fe + S \rightarrow FeS + \text{Heat}$. It is irreversible, forms a new stoichiometric compound, and releases significant chemical energy.